

On the formation of striations during fatigue crack growth

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Abstract: A study of the fatigue crack growth process in a low carbon steel has been carried out over a wide range of crack growth rates in air and vacuum at $R = 0.05$ using the techniques of stereo-microscopy and scanning-transmission electron microscopy (STEM). Examination of the profiles of striations indicated that they are of the stair-case type, as originally proposed by Schivje. In this steel the striations were not resolvable below a growth rate of 10^{-4} mm/cycle. In the crack growth rate range from 10^{-4} mm/cycle down to 10^{-8} mm/cycle a dislocation cell structure developed whose dimension in the direction of fatigue crack growth was approximately 1.5×10^{-4} mm/cycle, independent of the ΔK level. The constancy of this dimension has been interpreted to indicate that the fatigue crack growth process at low crack growth rates is discontinuous rather than cycle-by-cycle in nature. However the evidence gathered in this investigation supports the view that the fatigue crack growth process occurs on a cycle-by-cycle basis even at low fatigue crack growth rates.

As is known, striations are difficult to find on the fracture surfaces of specimens tested in vacuum. The use of the STEM technique provides evidence that a striation is created in vacuum on unloading, but that due to the greater extent of crack tip blunting in vacuum as compared to in air, in the next loading cycle the previously created striation is swept into the flanks of the blunted crack and largely eliminated.

For comparison purposes an aluminum alloy, 2519-T87, was also tested.

Key words: *Fatigue striations, Fatigue crack growth, Environmental effects, 1018 steel, 2519-T87 aluminum alloy*

1. INTRODUCTION

The process of fatigue crack growth has been studied rather extensively over the past fifty years, and although much has been learned about the process, there is still interest in the mechanism of striation formation and in whether or not a fatigue crack propagates in a cycle-by-cycle manner in the near threshold range. At growth rates of the order of 10^{-3} mm/cycle Forsyth and Rider [1] presented convincing evidence for cycle-by-cycle growth, and the crack tip plastic-blunting and re-sharpening model for fatigue crack growth developed by Laird and Smith [2] certainly involves this aspect. The early studies of striation spacings at lower crack growth rates involved the replica technique which has a limit of resolution of the order of 2×10^{-5} mm. Hertzberg and Paris [3] observed agreement between the striation spacings and the average crack growth rate down to 5×10^{-5} mm/cycle in 2024-T3 aluminum alloy, however there was considerable scatter in the striation spacings, and in establishing an average striation spacing the average of ten replicas was used. Broek and van der Vet [4] obtained a similar correlation in this same alloy down to 3×10^{-5} mm/cycle, but at lower crack growth rates the striation spacing ex-

ceeded the average growth rate, a circumstance attributed by them to non-uniform growth along the crack front at low ΔK levels. Broek later observed a correlation between striation spacings and crack growth rates down to 4×10^{-5} mm/cycle in both 2024-T3 and 7075-T6 [5]. In this early work on aluminum alloys there was general agreement that a fatigue crack advance in a cycle-by-cycle manner. Schivje [6] in fact concluded that "crack extension in principle occurs in every load cycle as a consequence of cyclic slip occurring in every load cycle. This also applies to microcracks for which growth lines have not been observed". More recently, Riemelmoser et al. [7] have presented an argument for cycle-by cycle propagation of fatigue cracks at small stress intensity ranges and attribute a minimum value of the striation spacing with decreasing amplitude of stress intensity to a limiting spacing between slip lines. However, that a fatigue crack should grow cycle-by-cycle at low growth rates is not now a universal view, for evidence has been presented which purports to show that in steel the growth process is intermittent in nature below a growth rate of 2×10^{-4} mm/cycle [8].

The objectives of the present study are:

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1. to examine the mechanism of striation formation in a low carbon steel, and

2. to determine whether or not cycle-by cycle fatigue crack growth occurs in this steel below a growth rate of 2×10^{-4} mm/cycle.

2. EXPERIMENTS

Fatigue crack growth studies were carried out at $R = 0.05$ in air and in vacuum using 6.35 mm thick compact specimens which had been machined from a plate of normalized 1018 steel. The aluminum alloy 2519-T87 was also examined for purposes of comparison. Thin foils containing portions of the fracture surfaces were examined in a high resolution STEM at 60 kV to obtain images of the fracture surfaces as well as of the underlying dislocation arrays. The stereo-imaging technique was used to analyze the details of the crack profile as a function of the ΔK level, and the sub-fracture surface dislocation arrays were examined by transmission electron microscopy. Crystallographic orientations were determined using electron diffraction and etch pitting techniques.

3. RESULTS

3.1. 1018 steel. The fatigue crack growth rate in 1018 steel as a function of ΔK at $R = 0.05$ in air and in vacuum is shown in Fig. 1.

The appearance of the striations created at growth rates above 2×10^{-4} mm/cycle in air is shown in Fig. 2a, and Fig. 2b shows the underlying dislocation microstructure. Examination of the striations by means of the stereo-imaging technique showed the striations were of the type shown in Fig. 3, known as the staircase (or saw-tooth) type. The bright lines in Fig. 2a were created on unloading, as were the narrow dislocation bands shown in Fig. 2b. The transition from loading to unloading portions of the striations were rounded, with opposite sides of the fracture surface mirroring each other. This type of profile has been observed by Schivje [6], Hertzberg [9], McMillan and Pelloux [10], Mughrabi et al. [11], Pelloux [12], and Bowles and Broek [13]. Electron diffraction studies of the fracture surfaces indicated the average crack growth plane to be $\{110\}$, the resultant of simultaneous slip on two $\{112\}$ in $\langle 111 \rangle$. The present results are only for $R = 0.05$ loading, but it is noted that Murakami et al. [14] have observed that the height of the staircase profile is a function of the R ratio, being less for $R = 0.1$ than for $R = 0.5$.

Figure 4 shows the geometrical characteristics

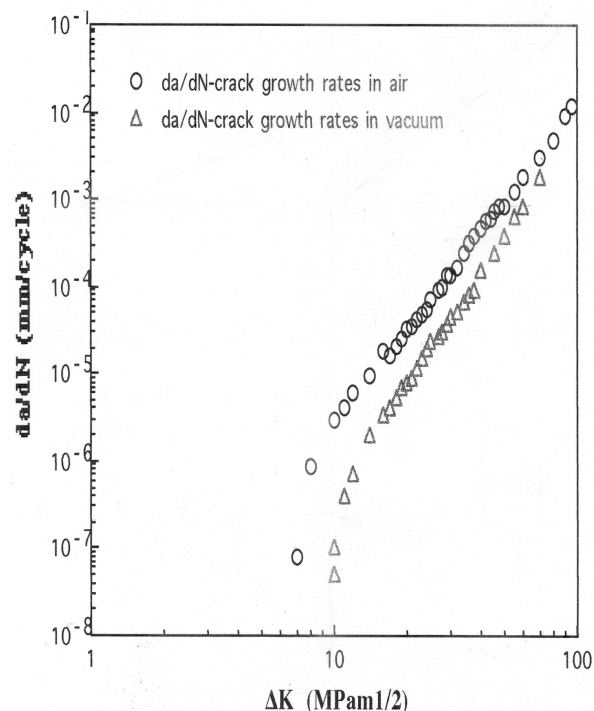


Fig. 1. The rate of fatigue crack growth in 1018 steel in air and in vacuum. ($R = 0.05$).

of the striations as determined by measurements of stereo-pairs as a function of the ΔK level.

The appearance of the fatigue fracture surface which developed at crack growth rates below 1.5×10^{-4} mm/cycle is shown in Fig. 5a. It is characterized by heavy striation-like markings spaced at an interval of about 1.5×10^{-4} mm. The area fraction containing this type of marking increased rapidly below a growth rate of 10^{-4} mm/cycle, reaching a level of 90% at a growth rate of 10^{-6} mm/cycle. The underlying dislocation array which was developed in the low growth rate regime is shown in Fig. 5b. It consists of a cell structure which is elongated in the direction parallel to the crack front, which was usually $\langle 111 \rangle$, with the rotation across a cell boundary being about 5° . It is significant that the width of a cell in the direction of fatigue crack propagation is 1.5×10^{-4} , exactly the same spacing as that of the striation-like markings. It is further noted that these striation-like markings were present even at the threshold level, as was the underlying cell structure. The spacing of the striation-like markings is indicated in Fig. 6a. Similar results obtained by Roven et al. [8] were interpreted to indicate that the fatigue crack grew in a discontinuous manner when striation-like markings

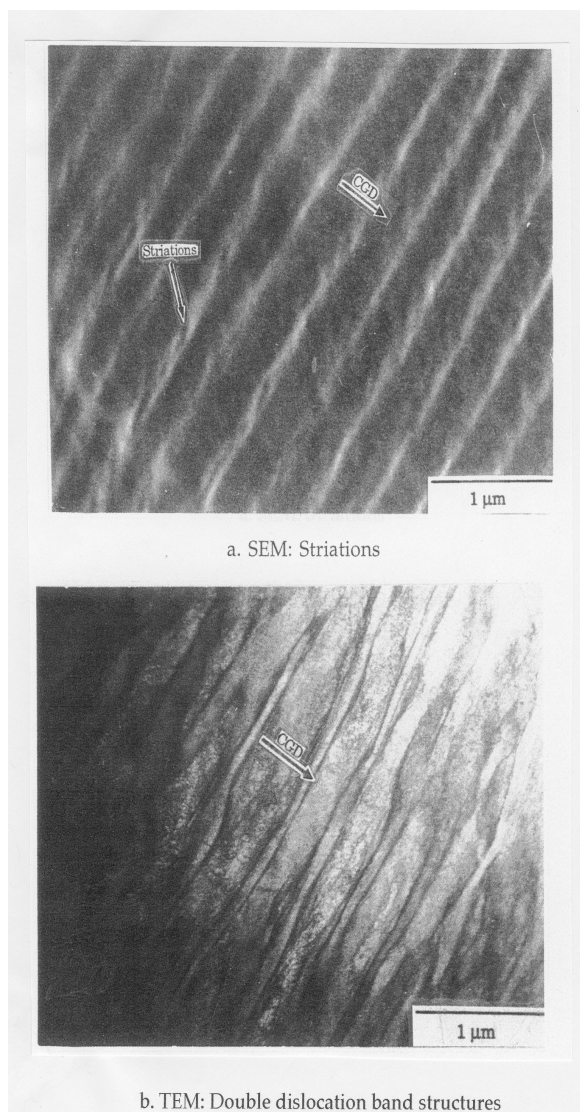


Fig. 2. Striations and dislocation bands in 1018 steel.

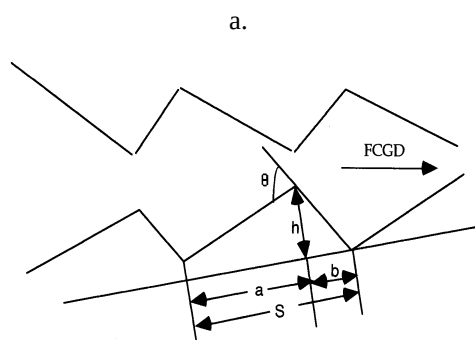


Fig. 4. Characteristics of striations in 1018 steel. a. Schematic showing characteristic dimensions. b. Striation dimensions a , b and h as a function of the striation spacing, S .

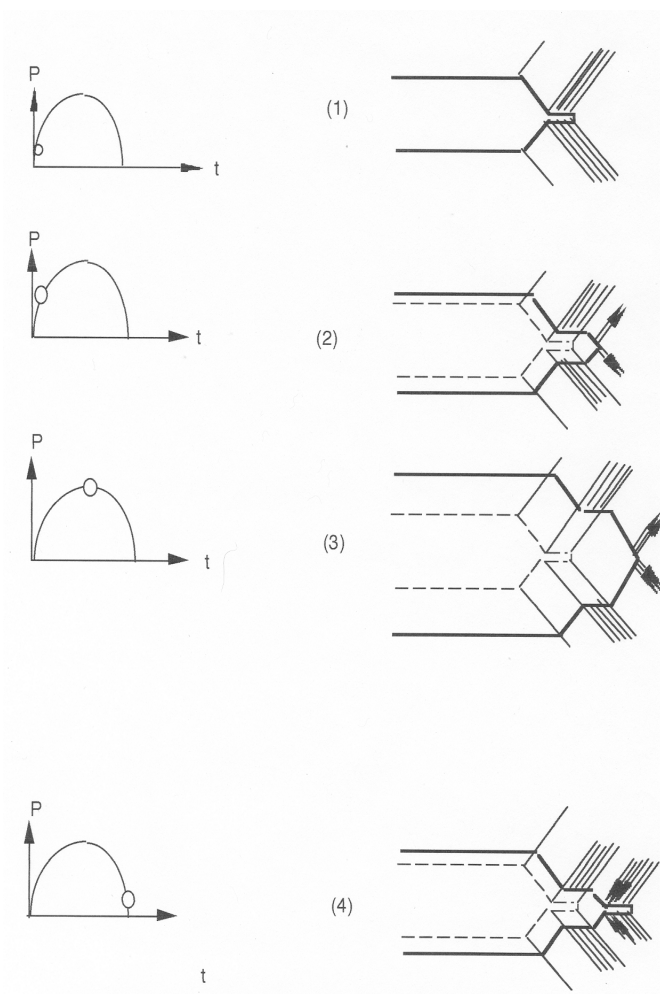
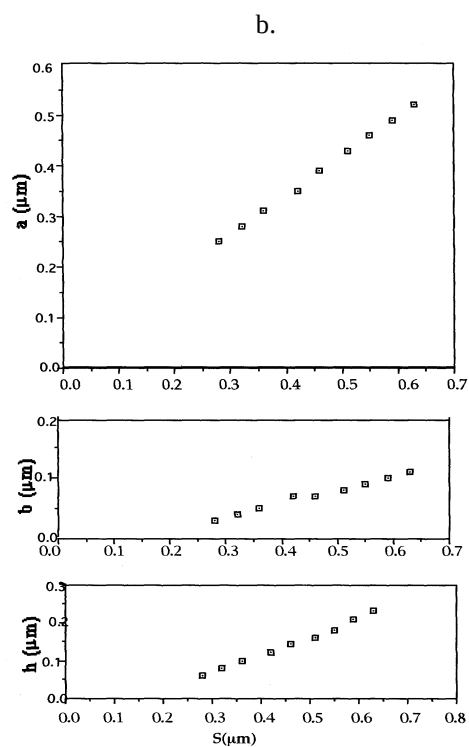


Fig. 3. Schematic model of staircase type of striation formation.



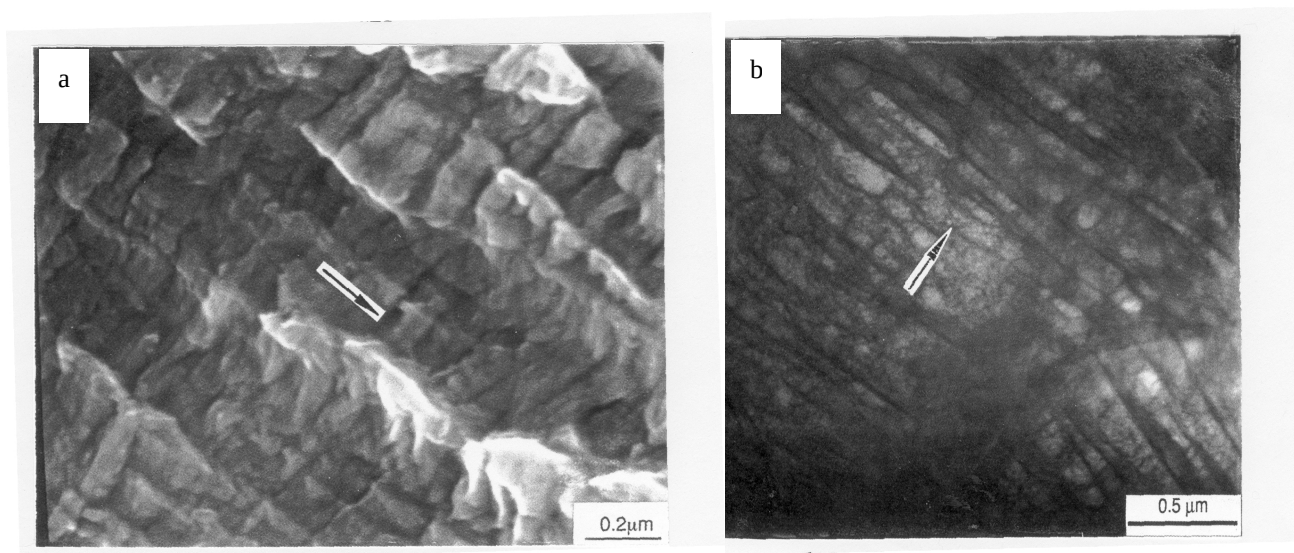


Fig. 5. a. Striation-like markings observed in 1018 steel observed at growth rates below 10^{-4} mm/cycle.
b. Dislocation cell structure underlying striation-like markings.

were formed. However, examination of the region between two striation-like markings created at crack growth rates in the interval from 10^{-4} to 10^{-5} mm/cycle revealed the presence of additional markings as shown in Fig. 5a. These markings were parallel to the striation-like markings, and their spacing closely corresponded to the corresponding crack growth rate, as shown in Fig. 6b.

In contrast to the higher growth range, the opposing fracture surfaces were not mirror images in the low growth regime. Instead they were nestled together, an indication that only one slip system at a time was dominant in this range as compared to two simul-

taneously acting slip systems at growth rates above 2×10^{-4} mm/cycle. Over much of the fracture surface the average overall growth plane was still $\{110\}$, but, as observed by Dickson [15], between two striation-like markings a zig-zag oscillation about this plane using $\{112\}$ occurred, an indication the slip systems at the crack tip operated in sequence rather than simultaneously. The faceting of the fracture surface features caused by the alternate use of $\{112\}$ planes is an indication that the fatigue crack did not propagate in a discontinuous manner, leaping from one striation-like marking to the next.

The appearance of a fatigue fracture surface

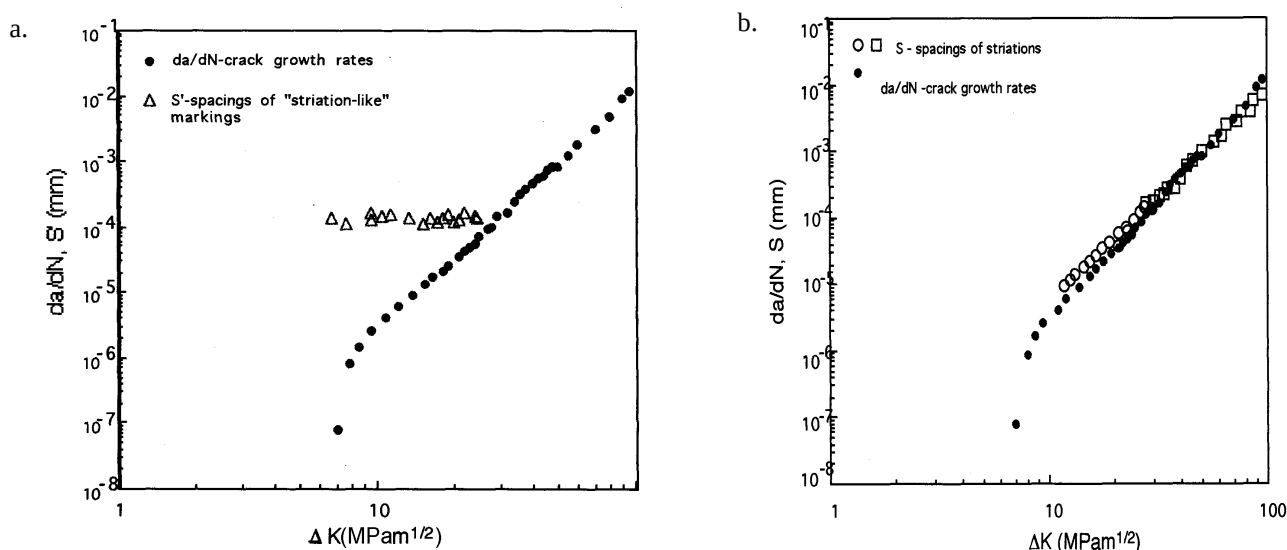


Fig 6. Comparison of fatigue crack growth rate with fracture surface markings. a. Striation-like markings.
b. Striation markings

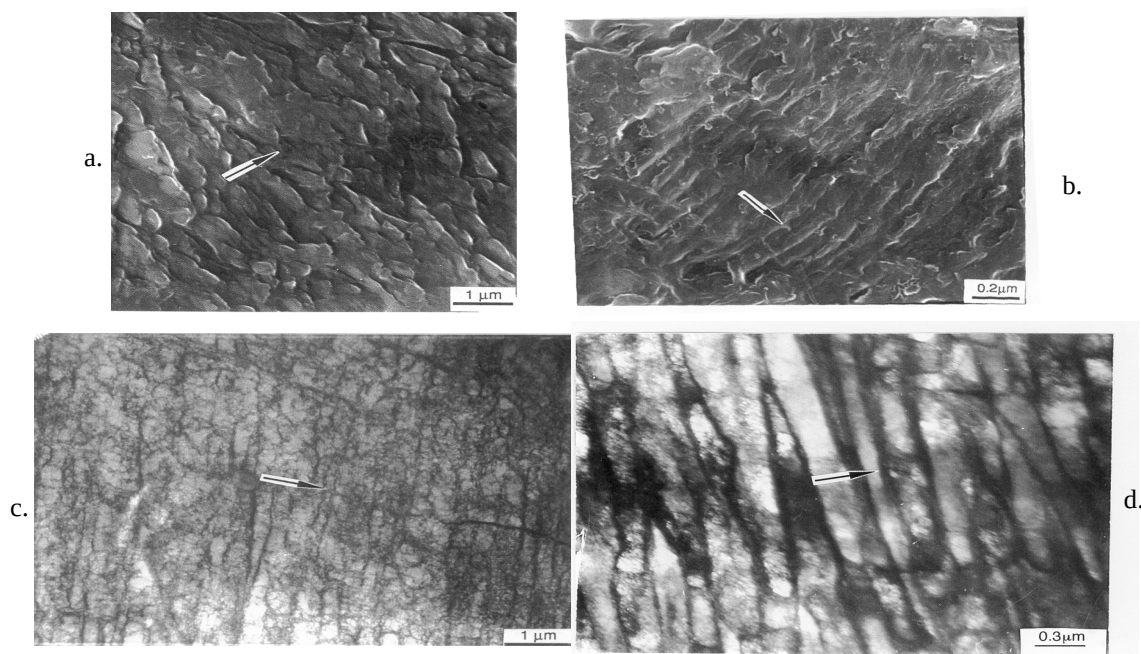


Fig. 7. 1018 steel tested in vacuum. a. and b. Fatigue fracture surfaces at $da/dN = 5 \times 10^{-4}$ mm/cycle and 10^{-7} mm/cycle, respectively. c and d. Dislocation structures underlying a. and b.

formed in vacuum at 5×10^{-4} mm/cycle is shown in Fig. 7a. It is quite irregular and no clear-cut correlation with the fatigue crack growth rate was made. The underlying dislocation arrays are shown in Fig. 7b, and a correlation can be made between the spacing of the dislocation bands and the rate of fatigue crack growth.

At a growth rate of 10^{-7} mm/cycle in vacuum striation-like markings were observed as in Fig. 7c. The underlying dislocation cell-structure is also shown in Fig. 7d.

The absence of striation markings in vacuum has been attributed to the comparatively extensive crack tip blunting that takes place in vacuum as compared to in air [16]. As indicated in Fig. 8, the ratio of the CTOD to crack advance is much higher in vacuum than in air. During the crack opening process in air the striation created upon the prior unloading is largely undisturbed. During the opening process in vacuum the striation which had been created in the previous unloading step is swept into the crack tip flanks, and as a result is largely eliminated by the extensive stretch involved. Evidence of the unloading event is still retained however within the dislocation structure (Fig. 7b).

3.2. Aluminum alloy 2519-T87 The appearance of fatigue striations formed in air at a growth rate of 10^{-3} mm/cycle in 2519 is shown in Fig. 9a. The underlying dislocation substructure is shown in Fig. 9c. At 10^{-7} mm/cycle the fracture appearance was faceted, as

shown in Fig. 9b, and the underlying dislocation structures were diffuse, i.e., no substructures were formed (Fig. 9d).

4. DISCUSSION

From these results we note the following.

a. In comparing the fractographic results obtained in air for 1018 steel and 2519 aluminum alloy above a crack

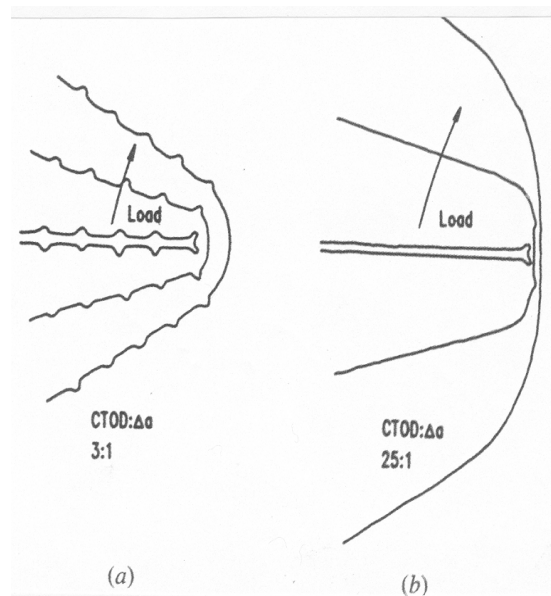


Fig. 8. Effect of extent of crack-tip blunting on striations. a. In air, striations remain. b. In vacuum, striations eliminated.

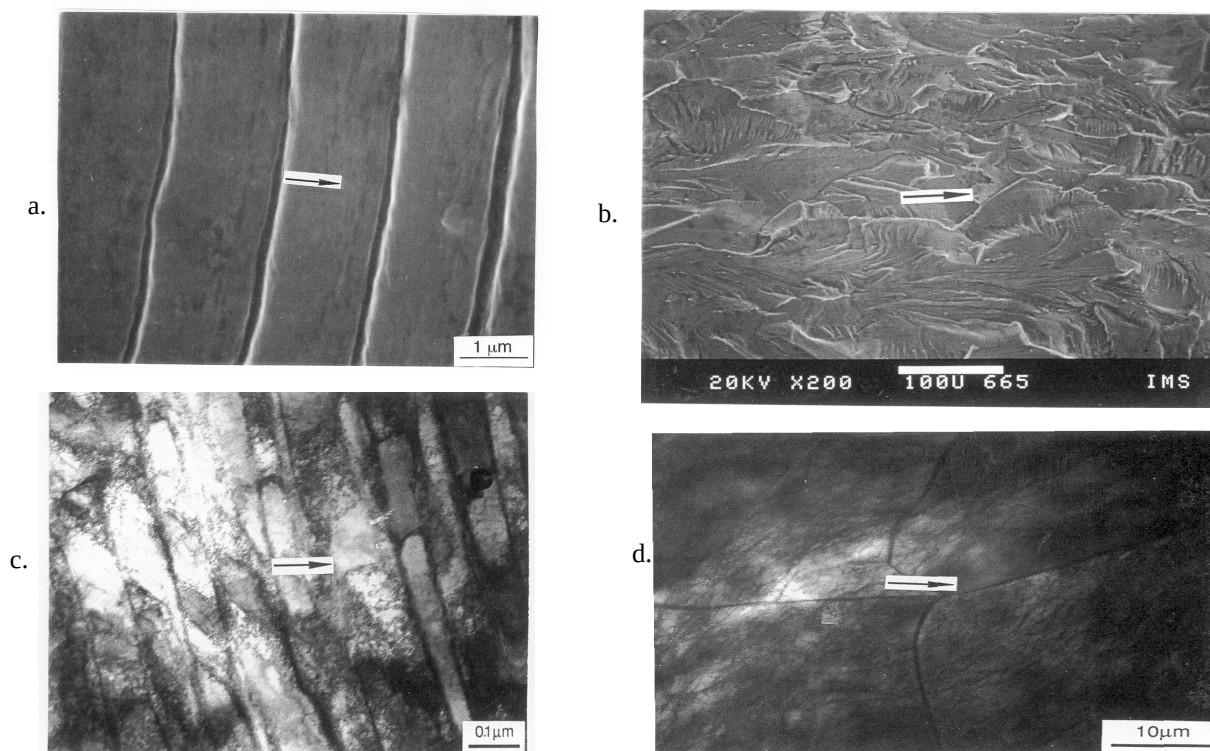


Fig. 9. Al 2519-T87. a. Striations at 10^{-3} mm/cycle. b. Fracture surface at 10^{-7} mm/cycle.

c. Dislocation structure underlying a.

d. Dislocation structure underlying b.

growth rate of 10^{-4} mm/cycle it appears that the environment contributed more to crack advance per cycle in the case of the aluminum alloy than for the steel. In 2519 crack advance occurred by a process of brittle fracture and blunting in each cycle, in accord with the model of Nix and Flower [17]. On the other hand in the 1018 steel the environment played a lesser role, and a saw-tooth-like crack profile developed. At growth rates above 10^{-4} the dislocation arrays created in both the steel and the aluminum alloy were strongly influenced by the unloading portion of a fatigue cycle.

b. At crack growth rates below 10^{-4} mm/cycle a dislocation substructure was formed in the 1018 steel but not in the aluminum alloy. The substructure observed in the steel at crack growth rates below 10^{-4} mm/cycle developed because dislocations have a greater ability to cross-slip and rearrange themselves into low energy cell structures in ferrite as compared to the aluminum alloy. In addition, the large number of cycles required per millimeter of crack growth allows a greater time for dislocation rearrangement into a cell configuration. This substructure led to the formation of striation-like markings on the steel fracture surface due to an interaction between the crack front and the cell walls, with the spacing of these striation-like markings being equal to the spacing of the cell walls in the direction of crack propagation. It is considered to be significant that no

such markings were observed in the aluminum alloy, nor did a substructure develop in the aluminum alloy.

c. The fact that striations were detected between the striation-like markings at fatigue crack growth rates between 10^{-4} and 10^{-5} mm/cycle indicates that the fatigue crack growth process occurred cycle-by-cycle in this range rather than on an intermittent basis, especially since the spacing of these markings corresponded to the average crack growth rate. Further, the general appearance of the fracture surface was unchanged from 10^{-4} to 10^{-7} mm/cycle, suggesting that cycle-by-cycle growth occurred below 10^{-4} mm/cycle.

5. CONCLUSIONS

1. The fatigue striations formed in air in 1018 steel at growth rates between 10^{-4} and 10^{-2} mm/cycle were of a staircase type.
2. Striations cannot be observed in 1018 steel at growth rates less than 10^{-4} mm/cycle.
3. In 1018 steel pronounced striation-like markings, spaced at a constant interval of 1.5×10^{-4} mm, were observed at all growth rates below 1.5×10^{-4} mm/cycle. These markings represent an interaction of the advancing crack front with a dislocation cell boundary.
4. Striation formation in the aluminum alloy 2519-T87 is much more sensitive to the environment than in 1018 steel. As a result the striations formed in the two ma-

terials are markedly different in appearance.

5. In vacuum, extensive blunting largely eliminates striation markings.

6. No evidence for fatigue crack growth on other than a cycle-by-cycle basis was found.

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